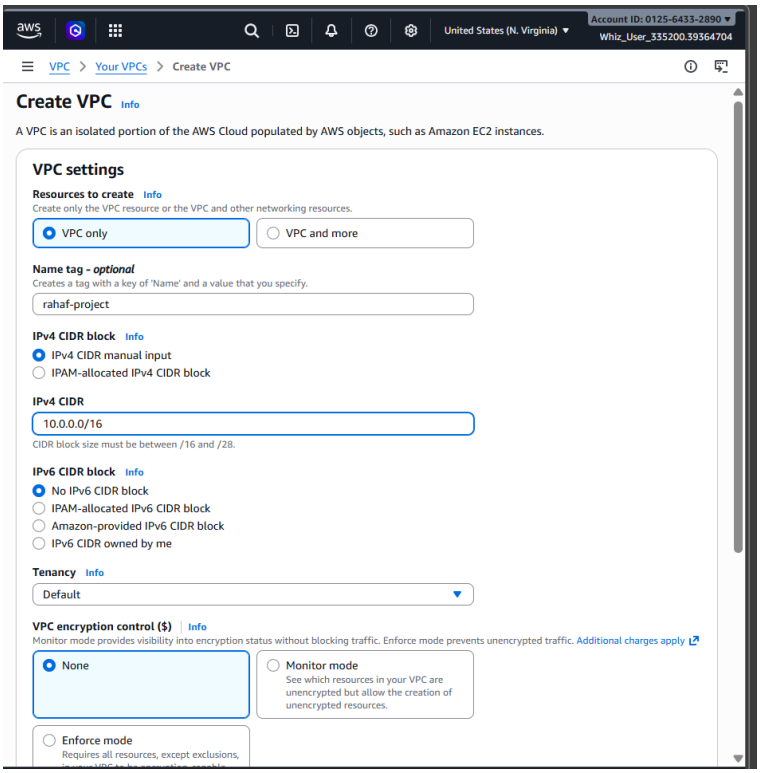
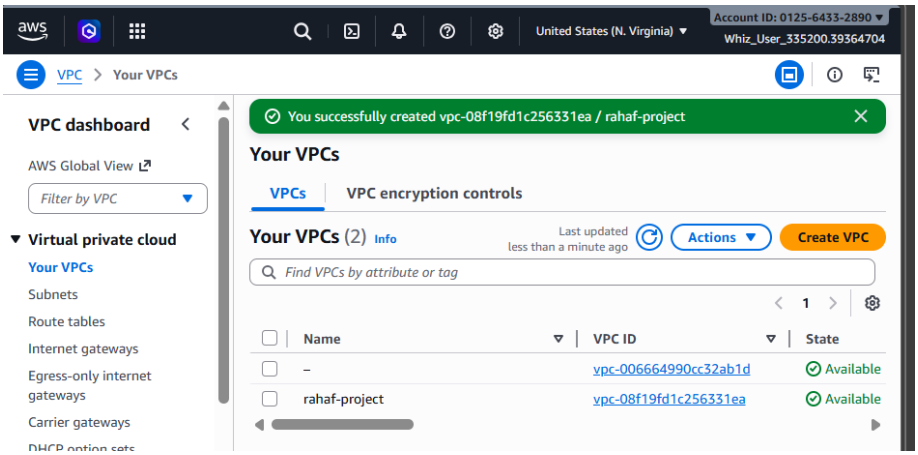
1. Networking Layer (VPC Design)

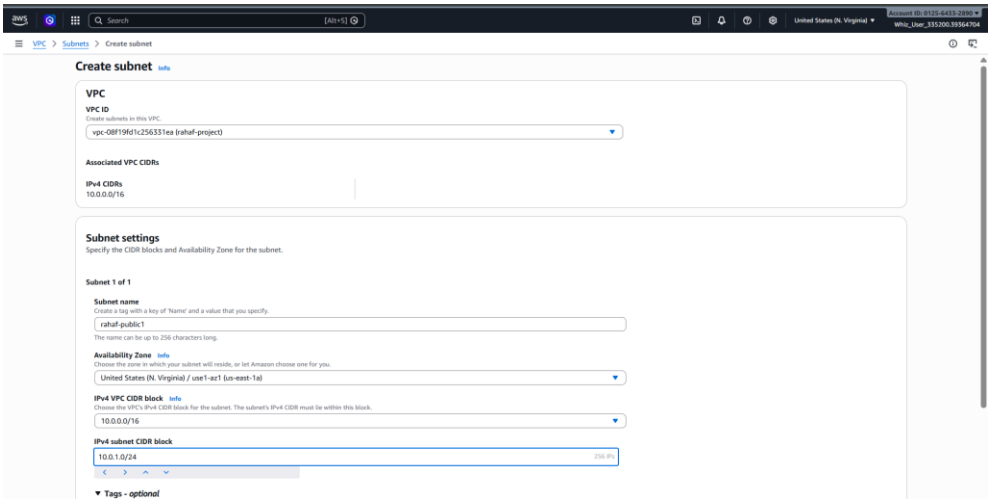
• Create a VPC

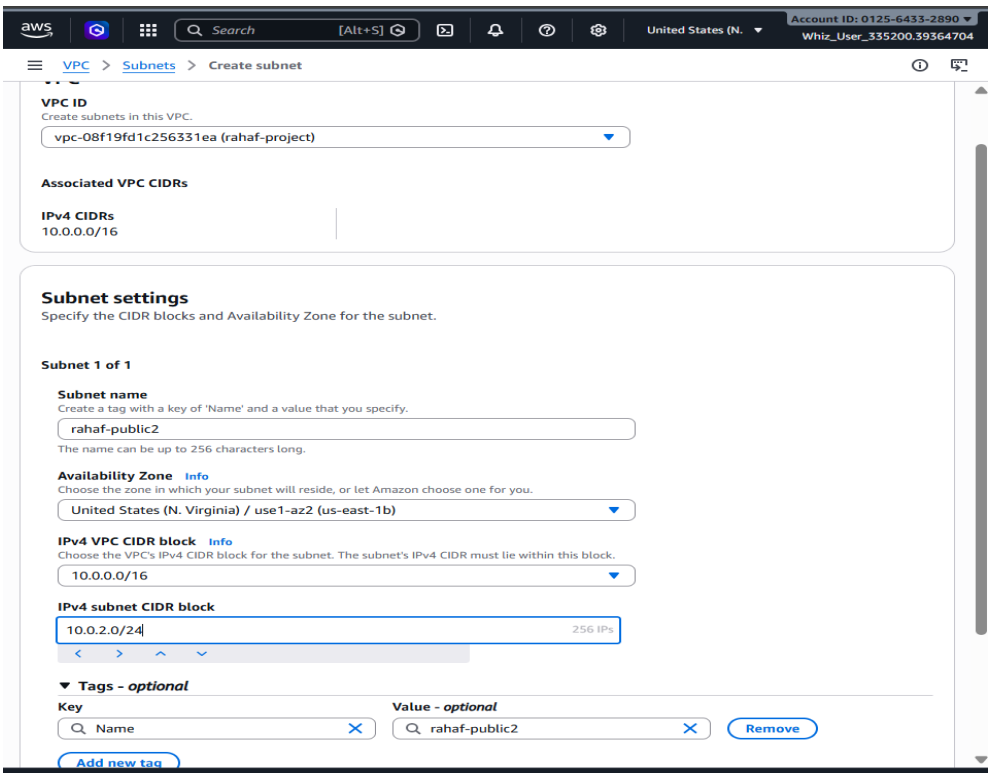




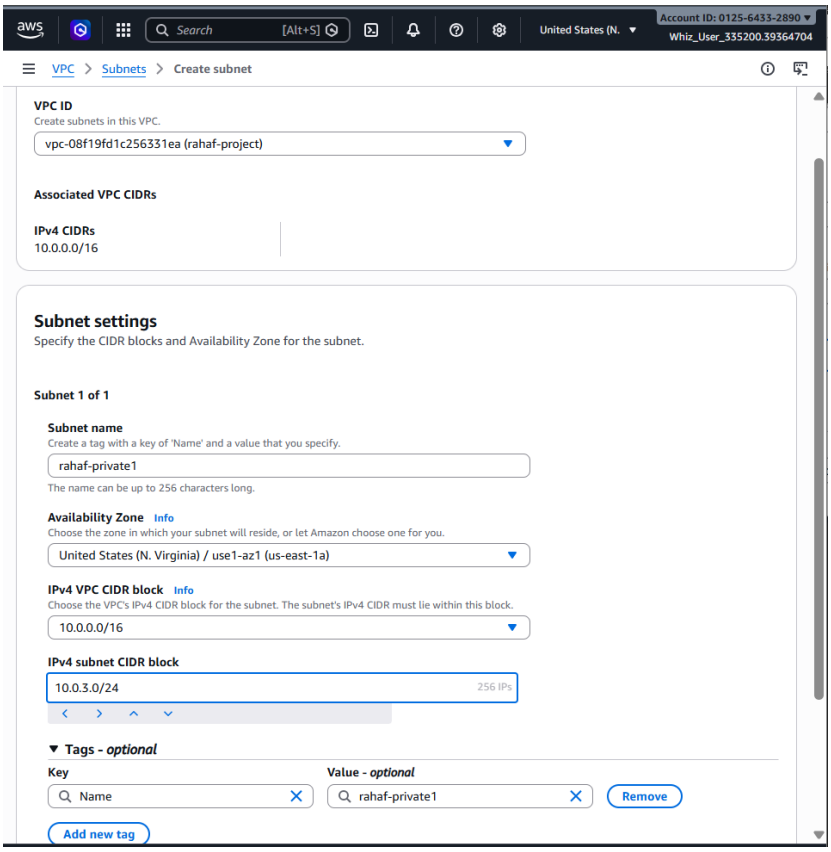
- **Create at least:**

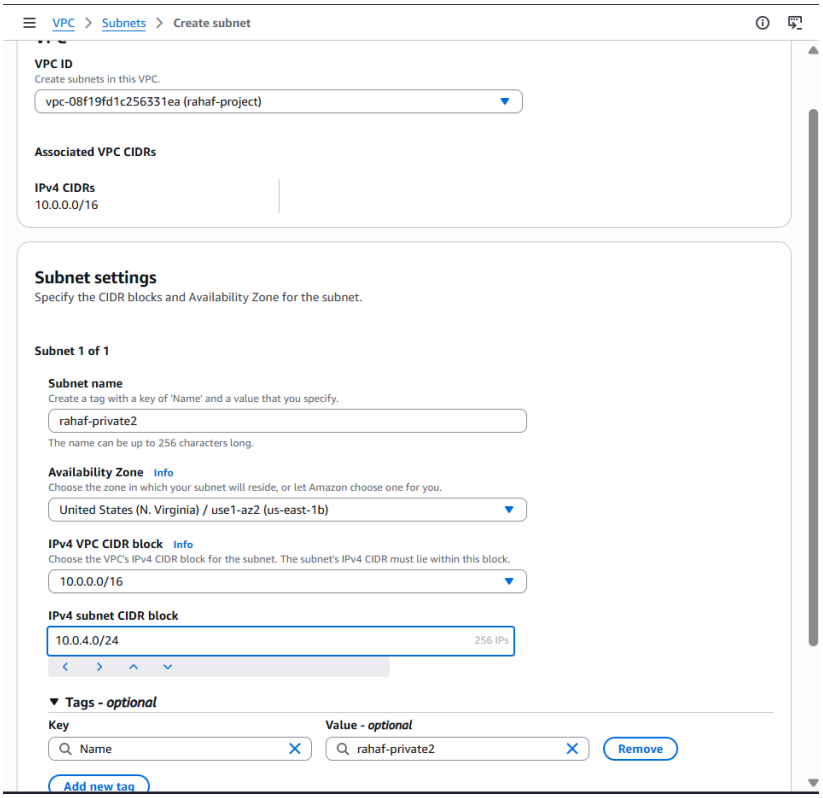
2 Public Subnets (different Availability Zones)

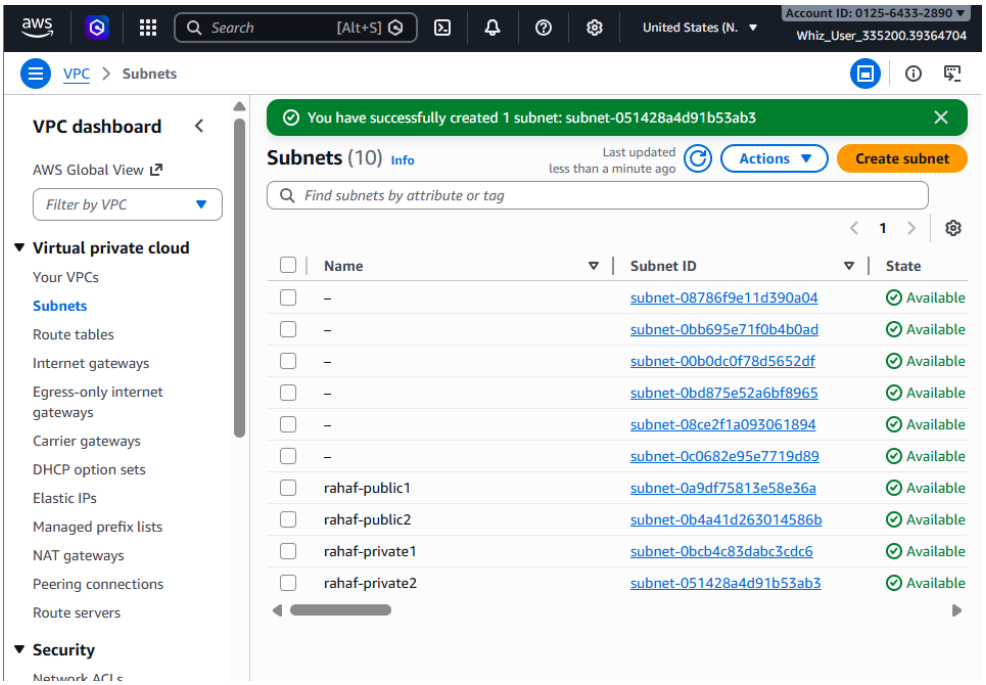




2 Private Subnets (different Availability Zones)

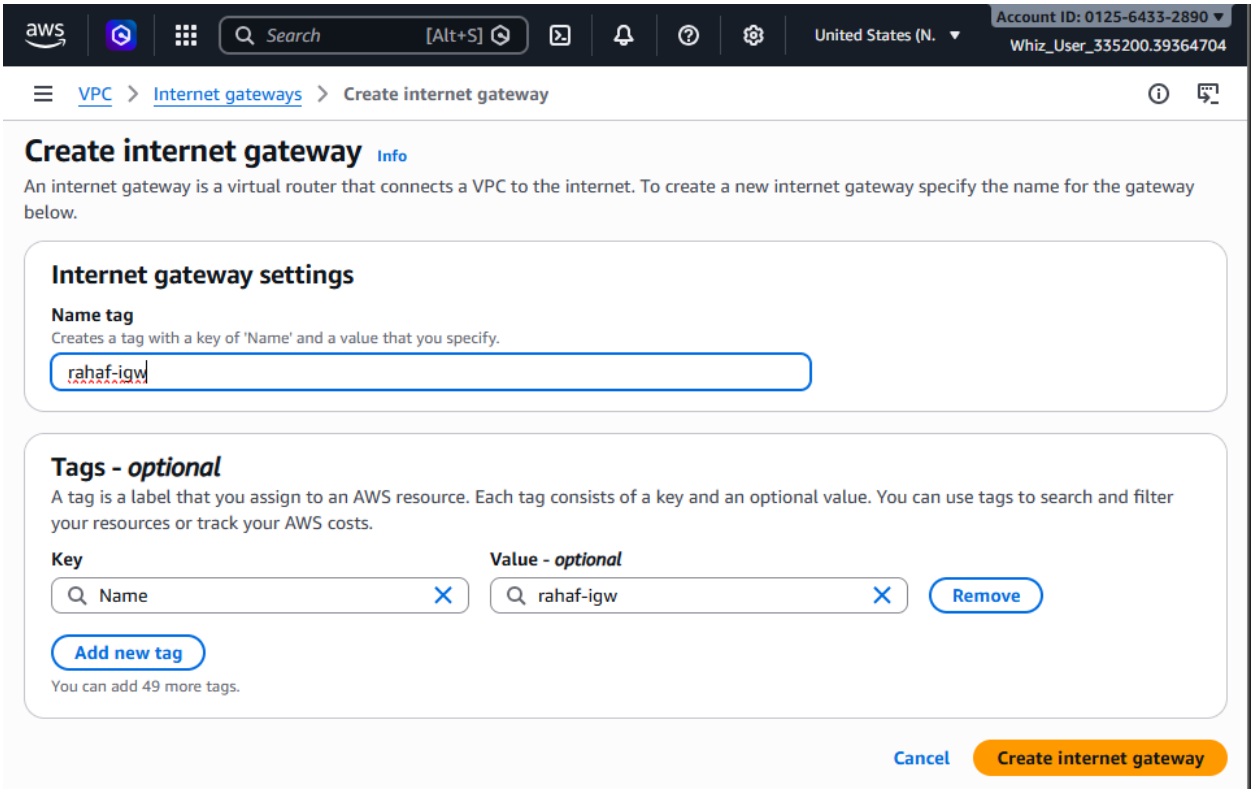


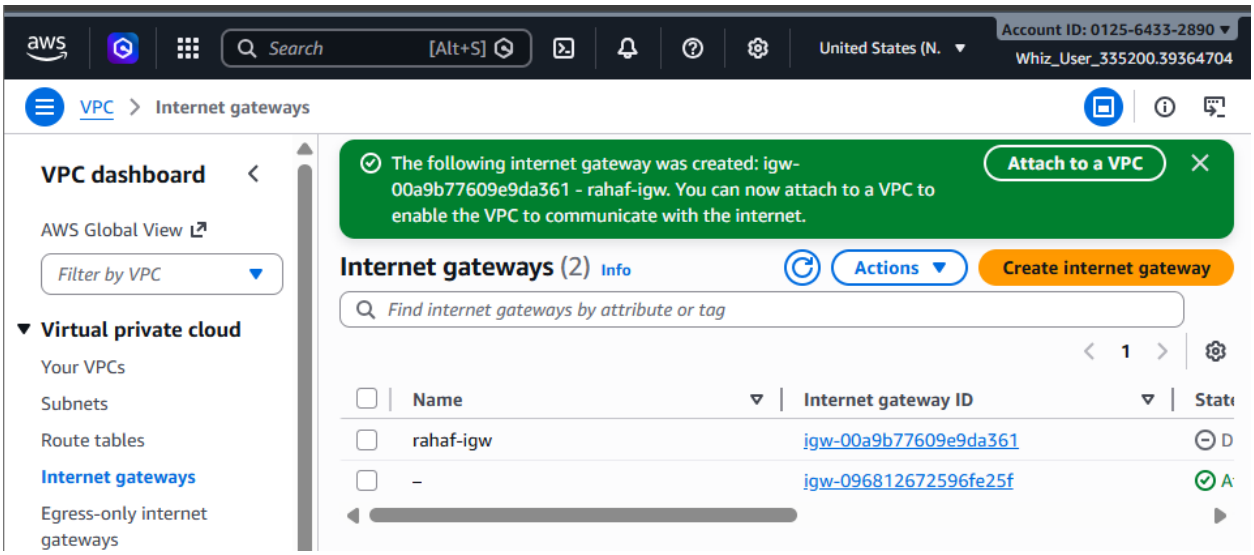


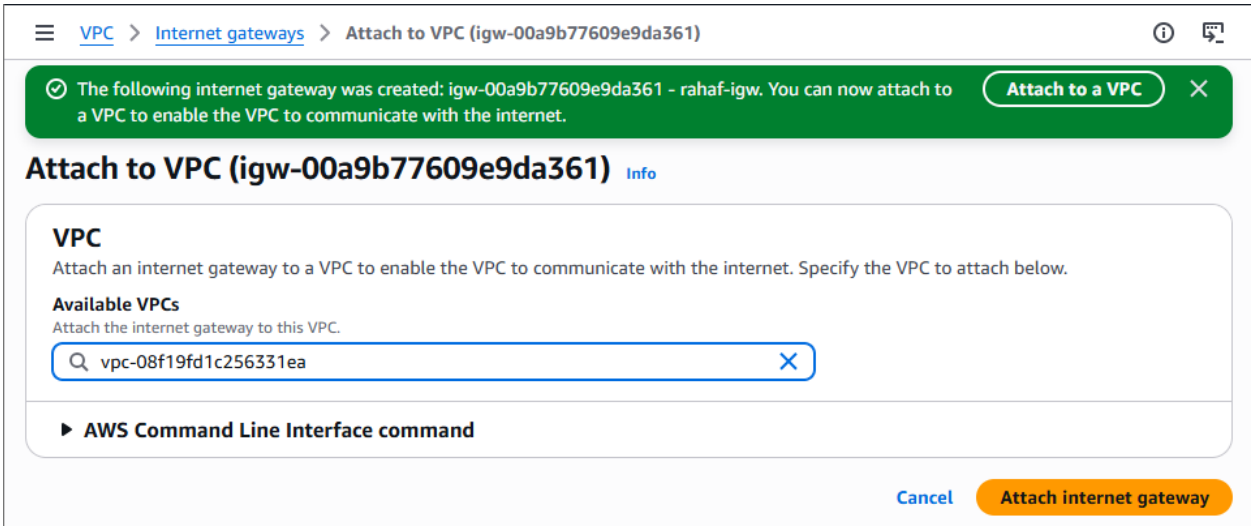


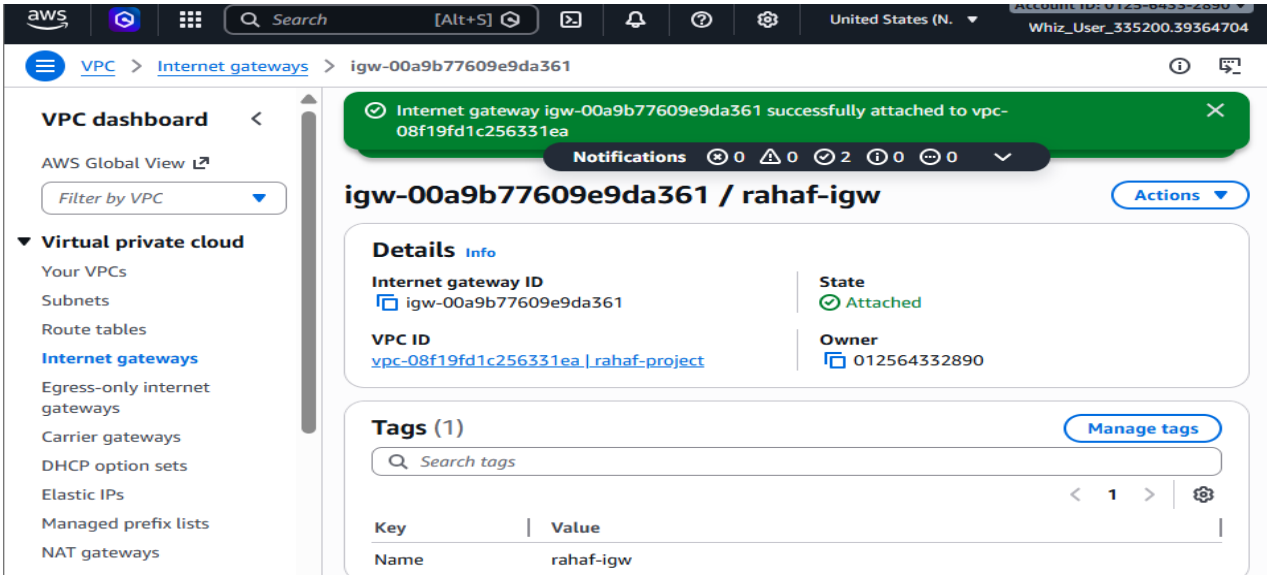
- **Configure:**

internet Gateway (IGW)

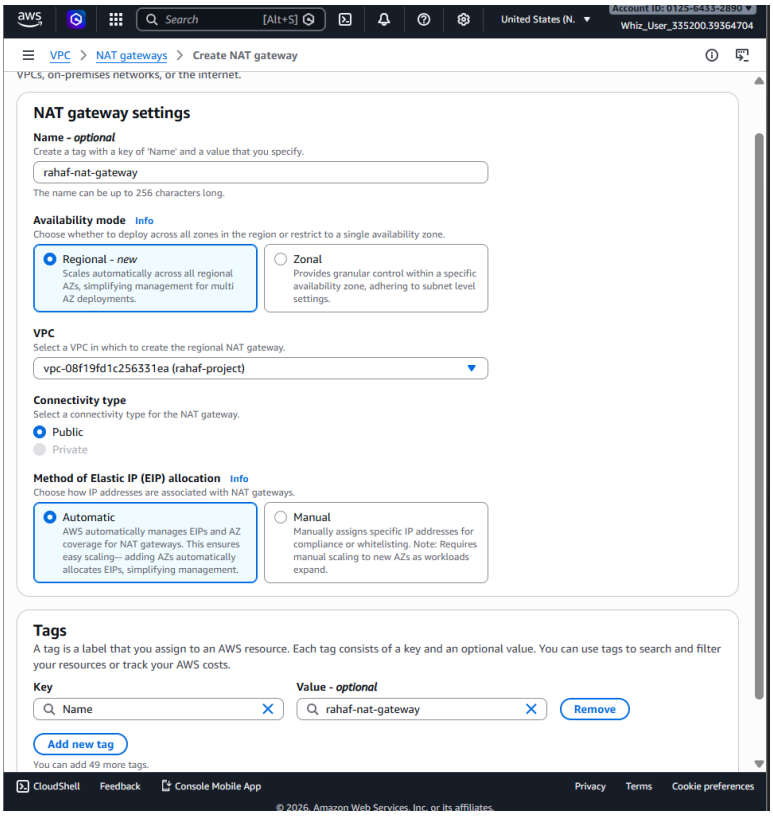


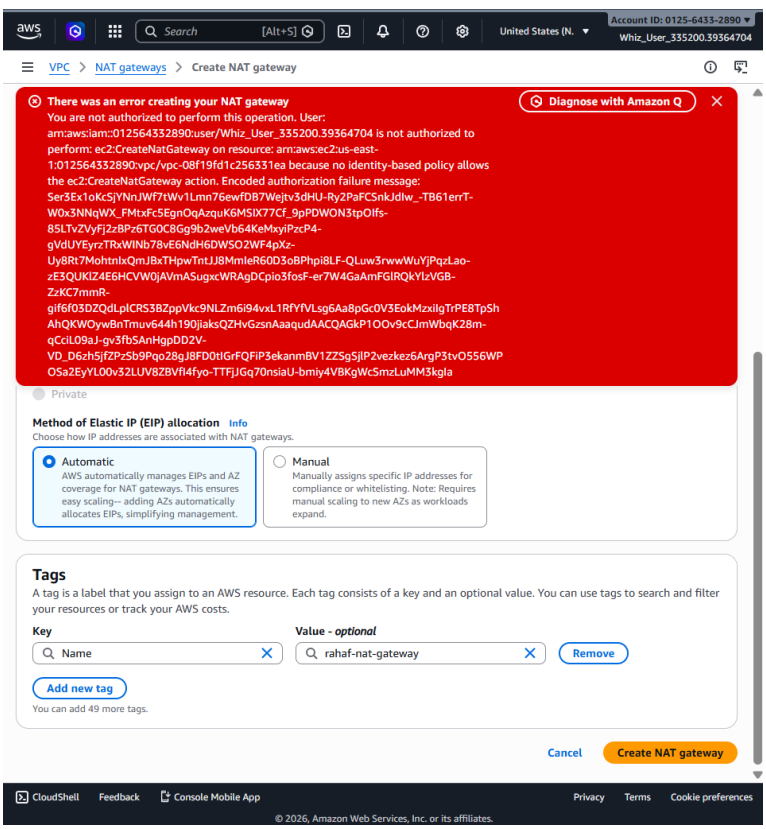

Search [Alt+S]
United States (N.
Account ID: 0125-6433-2890
Whiz_User_335200.39364704
VPC > Internet gateways
VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
The following internet gateway was created: igw-00a9b77609e9da361 - rahaf-igw. You can now attach to a VPC to enable the VPC to communicate with the internet.
Attach to a VPC
Internet gateways (2) Info
Actions
Create internet gateway
Find internet gateways by attribute or tag
Name
Internet gateway ID
rahaf-igw
igw-00a9b77609e9da361
–
igw-096812672596fe25f


VPC > Internet gateways > Attach to VPC (igw-00a9b77609e9da361)
The following internet gateway was created: igw-00a9b77609e9da361 - rahaf-igw. You can now attach to a VPC to enable the VPC to communicate with the internet.
Attach to a VPC
Attach to VPC (igw-00a9b77609e9da361) Info
VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.
Available VPCs
Attach the internet gateway to this VPC.
vpc-08f19fd1c256331ea
AWS Command Line Interface command
Cancel
Attach internet gateway


Search [Alt+S]
United States (N.
Whiz_User_335200.39364704
VPC > Internet gateways > igw-00a9b77609e9da361
VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Internet gateway igw-00a9b77609e9da361 successfully attached to vpc-08f19fd1c256331ea
Notifications
igw-00a9b77609e9da361 / rahaf-igw
Actions
Details Info
Internet gateway ID
igw-00a9b77609e9da361
State
Attached
VPC ID
vpc-08f19fd1c256331ea | rahaf-project
Owner
012564332890
Tags (1)
Manage tags
Search tags
Key
Value
Name
rahaf-igw

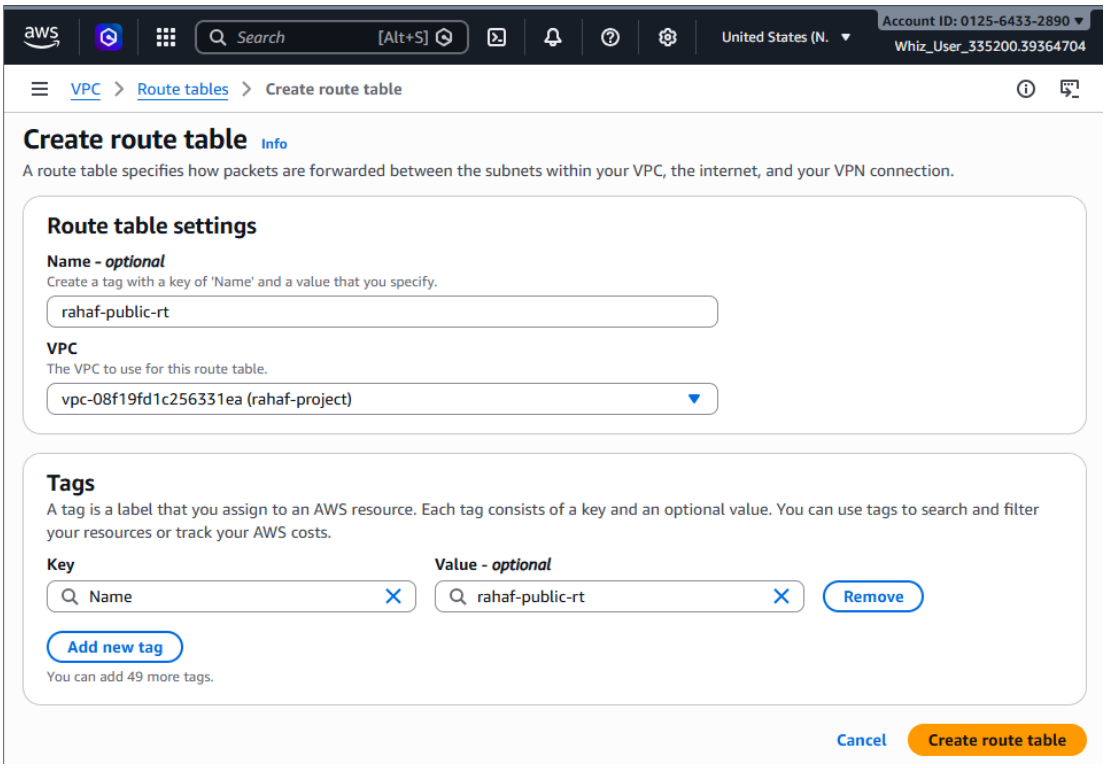
NAT Gateway

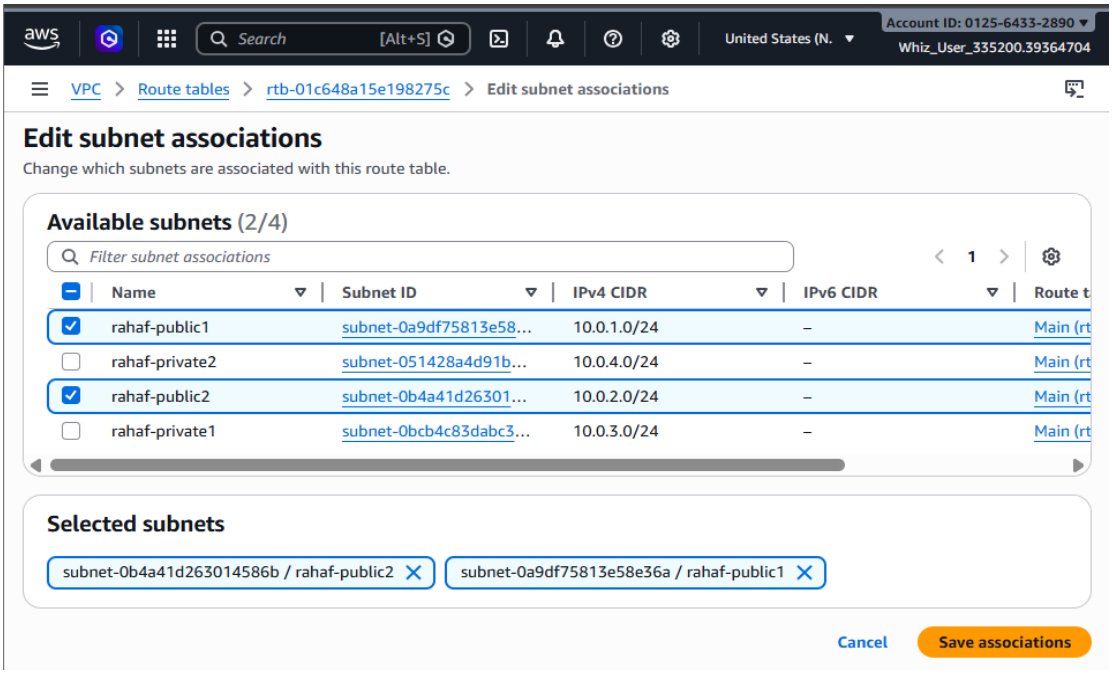

aws
Search
[Alt+S]
United States (N.
Account ID: 0125-6433-2890
Whiz_User_335200.39364704
VPC > NAT gateways > Create NAT gateway
VPCs, on-premises networks, or the internet.
NAT gateway settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
rahaf-nat-gateway
The name can be up to 256 characters long.
Availability mode Info
Choose whether to deploy across all zones in the region or restrict to a single availability zone.
Regional - new
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.
Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.
VPC
Select a VPC in which to create the regional NAT gateway.
vpc-08f19fd1c256331ea (rahaf-project)
Connectivity type
Select a connectivity type for the NAT gateway.
Public
Private
Method of Elastic IP (EIP) allocation Info
Choose how IP addresses are associated with NAT gateways.
Automatic
AWS automatically manages EIPs and AZ coverage for NAT gateways. This ensures easy scaling— adding AZs automatically allocates EIPs, simplifying management.
Manual
Manually assigns specific IP addresses for compliance or whitelisting. Note: Requires manual scaling to new AZs as workloads expand.
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Value - optional
Name
rahaf-nat-gateway
Remove
Add new tag
You can add 49 more tags.
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
© 2026, Amazon Web Services, Inc. or its affiliates.


aws
Search
[Alt+S]
United States (N.
Account ID: 0125-6433-2890
Whiz_User_335200.39364704
VPC > NAT gateways > Create NAT gateway
There was an error creating your NAT gateway
Diagnose with Amazon Q
You are not authorized to perform this operation. User: arn:aws:iam::012564332890:user/Whiz_User_335200.39364704 is not authorized to perform: ec2:CreateNatGateway on resource: arn:aws:ec2:us-east-1:012564332890:vpc/vpc-08f19fd1c256331ea because no identity-based policy allows the ec2:CreateNatGateway action. Encoded authorization failure message: Ser3Ex1oKcSjYNnJWf7tWv1Lmn76ewfDB7Wejtv3dHU-Ry2PaFCSnkJdlw_-TB61errT-W0x3NNqWX_FMtxFc5EgnOqAzquK6MSlX77Cf_9pPDWON3tpOlfs-85LTvZVyFj2zBPz6TG0C8Gg9b2weVb64KeMxyiPzcP4-gVdUYEyrzTRxWlNb78vE6NdH6DWSO2WF4pXz-Uy8Rt7MohtnIxQmJBxTHpwTntJJ8MmIeR60D3oBPhpi8LF-QLuw3rwwWuYjPqzLao-zE3QUKIZ4E6HCVW0jAVmASugxcWRAgDCpio3fosF-er7W4GaAmFGlRQkYIzVGB-ZzKC7mmR-gif6f03DZQdLplCRS3BZppVkc9NLZm6i94vxL1RfYfVLsg6Aa8pGc0V3EokMzxilgTrPE8TpShAhQKWOywBnTmuv644h190jiaksQZHvGzsnAaaqudAACQAGkP1OOv9cCJmWbqK28m-qCciL09aJ-gv3fbSAnHgpDD2V-VD_D6zh5jfZPzSb9Pqo28gJ8FD0tIGrFQFiP3ekanmBV1ZZSgSjlP2vezkez6ArgP3tvO556WPOSa2EyYL00v32LUV8ZBVfl4fyo-TTFjJGq70nsiaU-bmiy4VBKgWcSmzLuMM3kgla
Private
Method of Elastic IP (EIP) allocation Info
Choose how IP addresses are associated with NAT gateways.
Automatic
AWS automatically manages EIPs and AZ coverage for NAT gateways. This ensures easy scaling— adding AZs automatically allocates EIPs, simplifying management.
Manual
Manually assigns specific IP addresses for compliance or whitelisting. Note: Requires manual scaling to new AZs as workloads expand.
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Value - optional
Name
rahaf-nat-gateway
Remove
Add new tag
You can add 49 more tags.
Cancel
Create NAT gateway
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
© 2026, Amazon Web Services, Inc. or its affiliates.

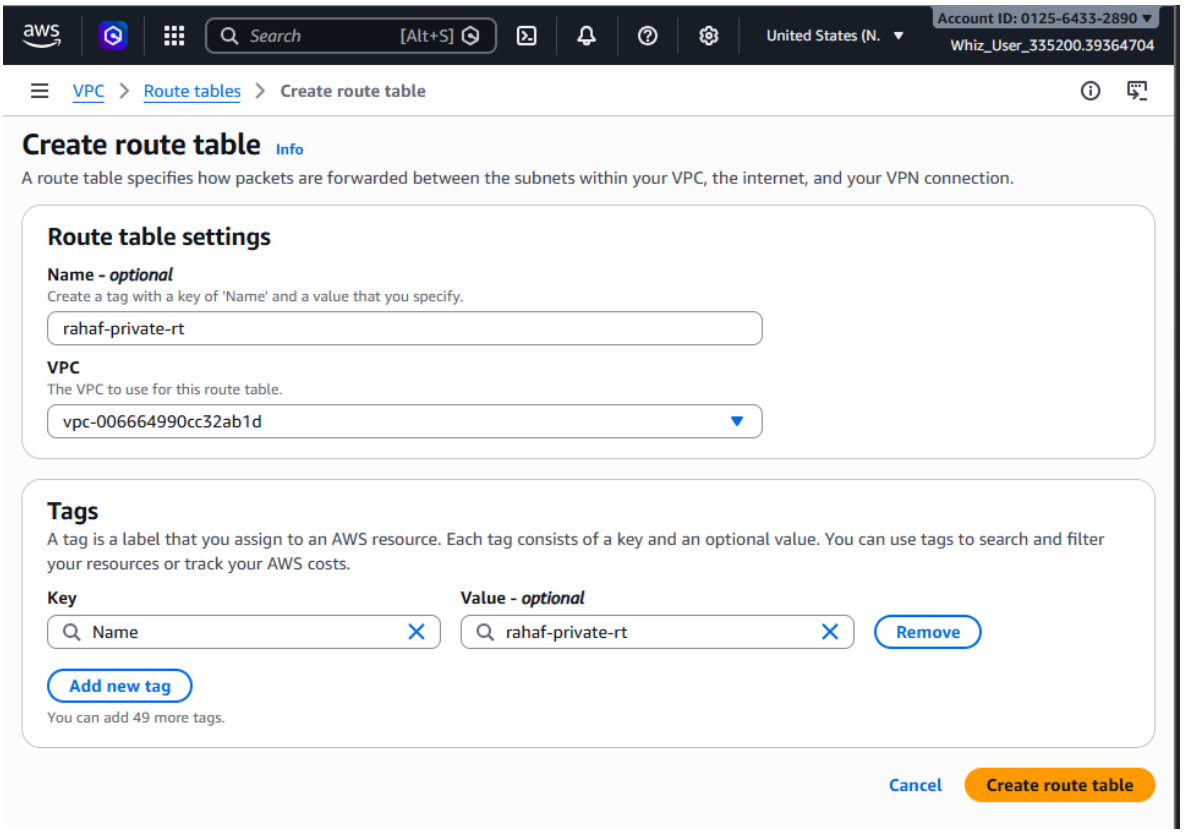
Route Tables

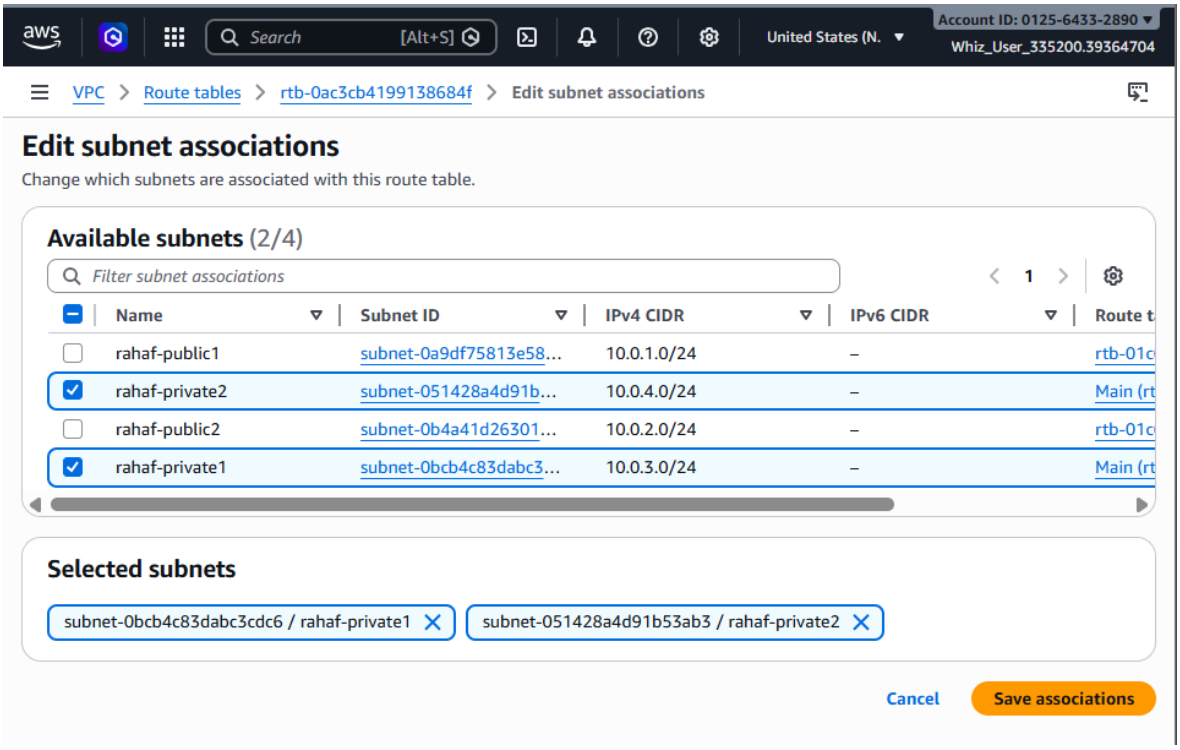
Here as you see I create public router table:





Here is a private router table

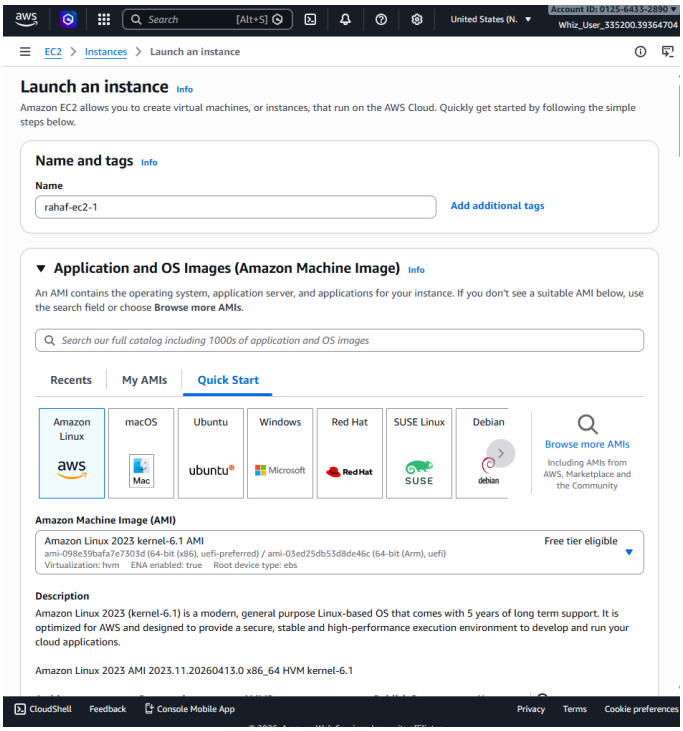




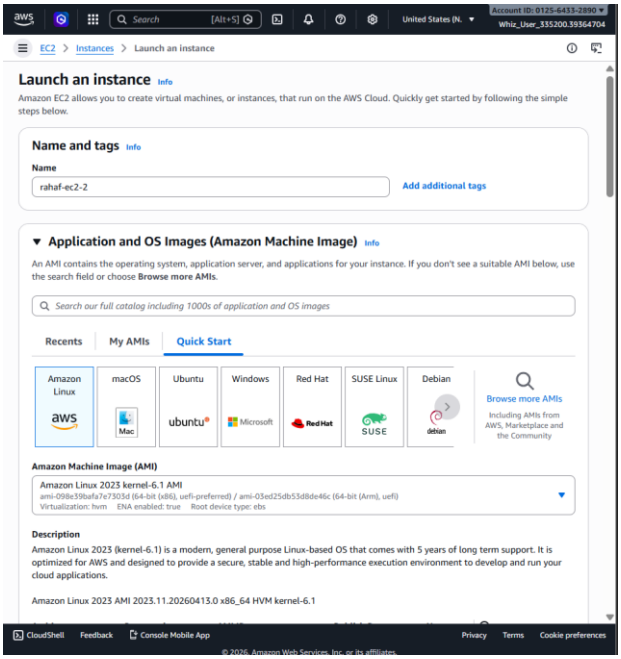
2. Compute Layer

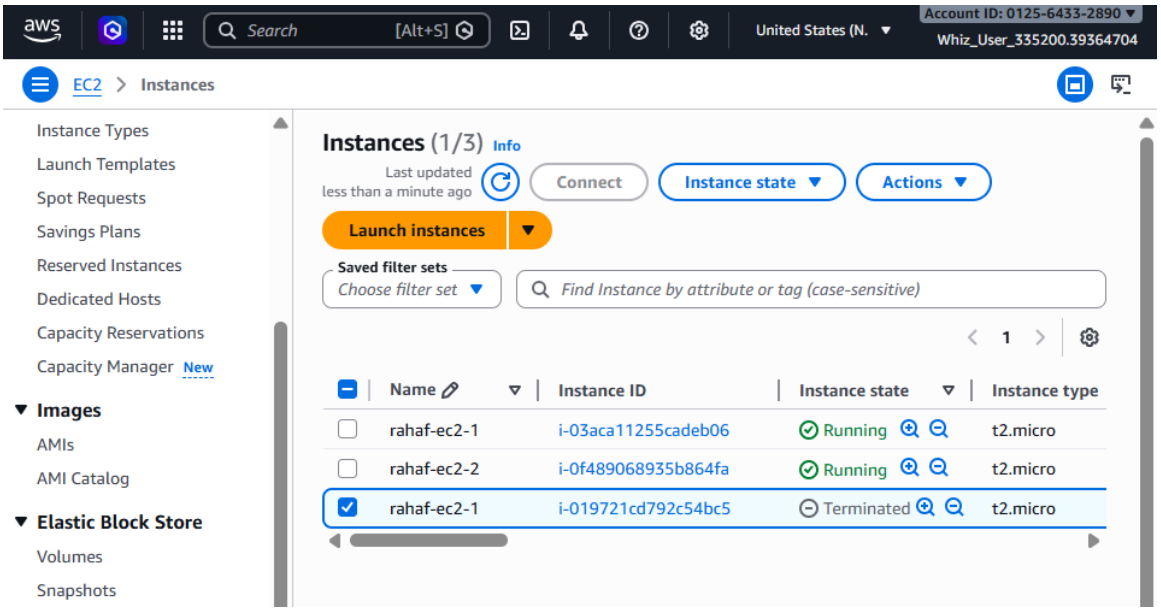
• Launch at least 2 EC2 instances in private subnets

EC2 1

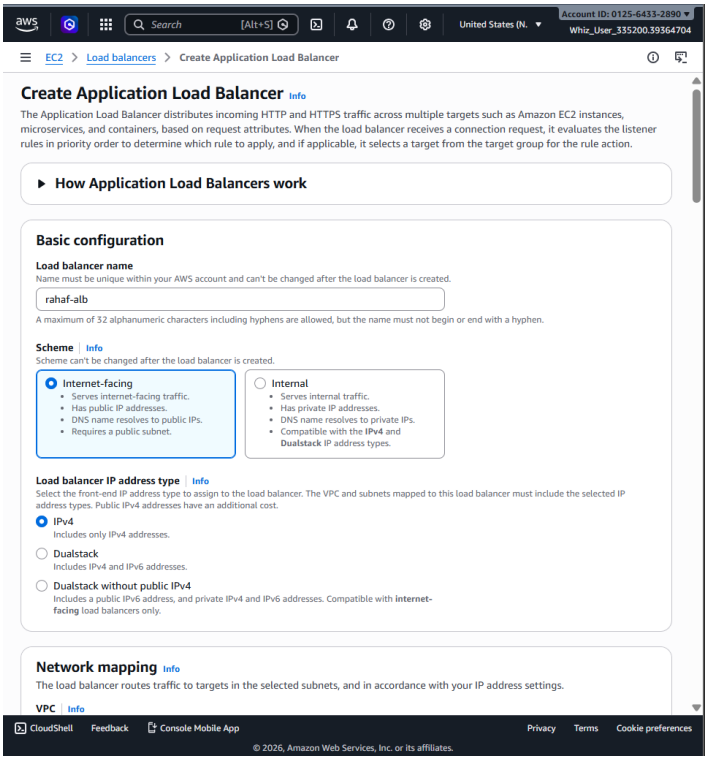


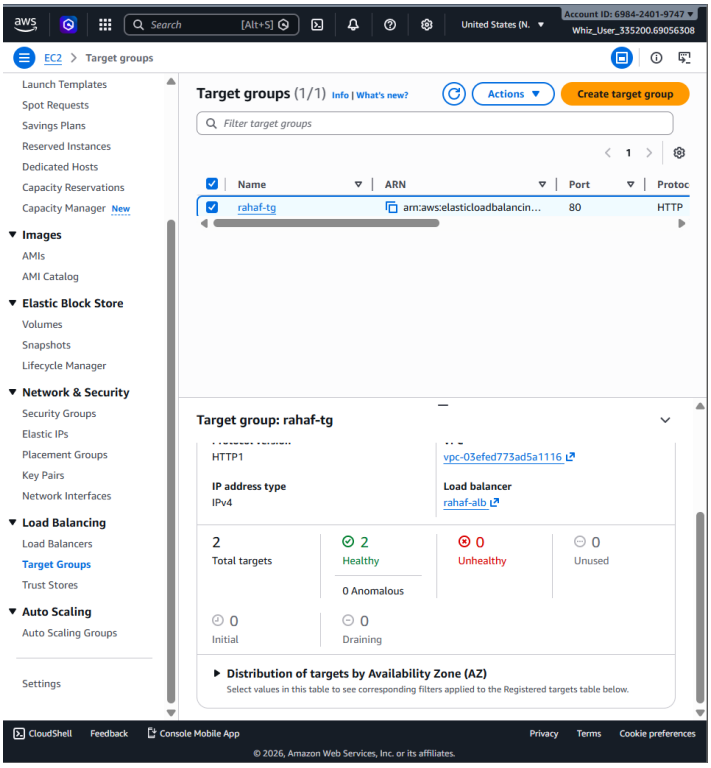
EC2 2



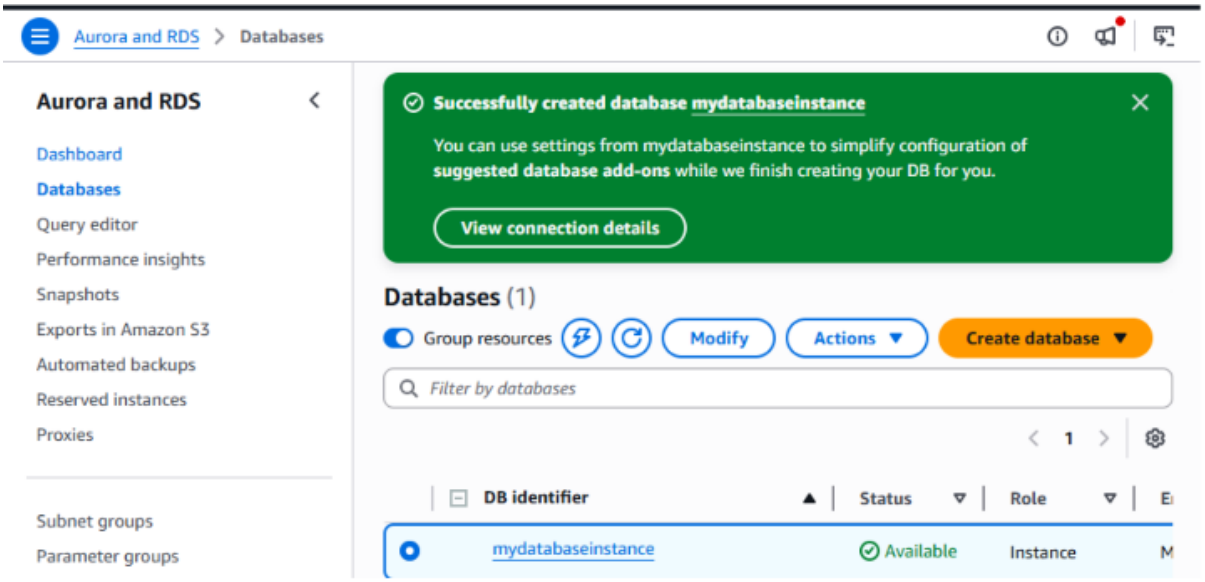


- **Ensure servers are reachable through Load Balancer only**



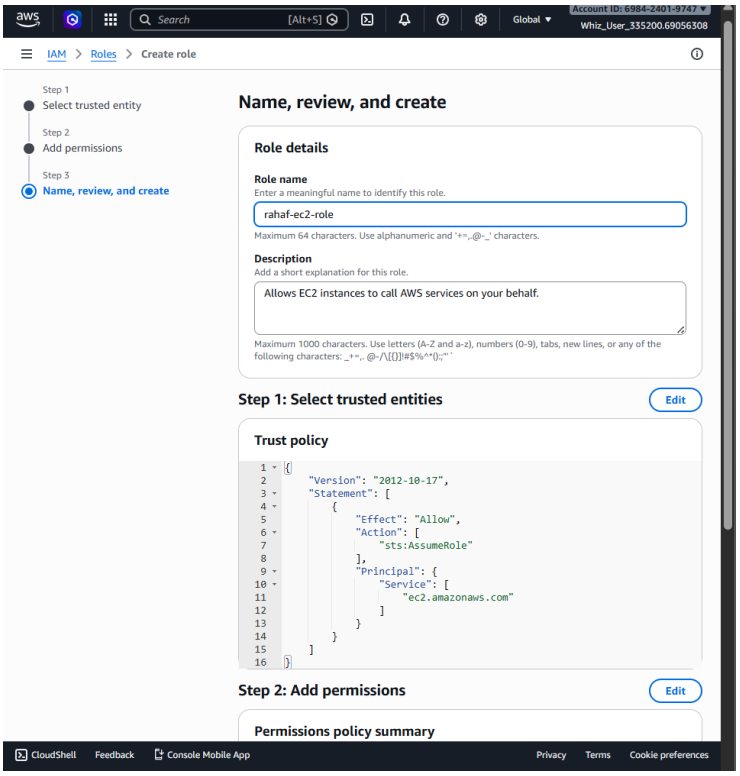

Account ID: 6984-2401-9747
Whiz_User_335200.69056308
EC2 > Target groups
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Target groups (1/1) Info | What's new?
Actions
Create target group
Filter target groups
Name
ARN
Port
Protoc
rahaf-tg
arn:aws:elasticloadbalancin...
80
HTTP
Target group: rahaf-tg
HTTP1
vpc-03efed773ad5a1116
IP address type
IPv4
Load balancer
rahaf-alb
2
Total targets
2
Healthy
0 Anomalous
0
Unhealthy
0
Unused
0
Initial
0
Draining
Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
© 2026, Amazon Web Services, Inc. or its affiliates.

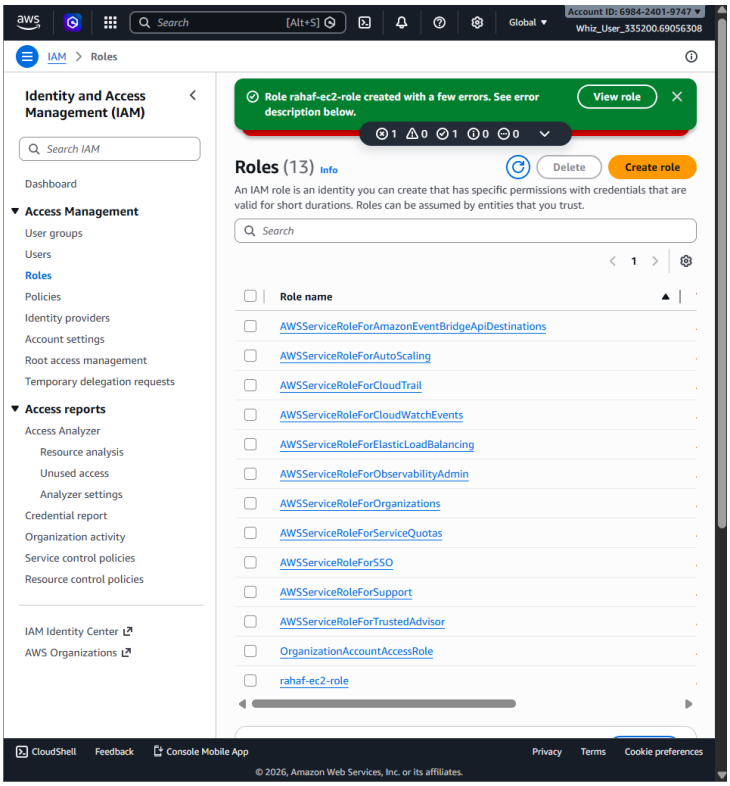
5. Data Layer


Aurora and RDS > Databases
Aurora and RDS
Dashboard
Databases
Query editor
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Successfully created database mydatabaseinstance
You can use settings from mydatabaseinstance to simplify configuration of suggested database add-ons while we finish creating your DB for you.
View connection details
Databases (1)
Group resources
Modify
Actions
Create database
Filter by databases
DB identifier
Status
Role
mydatabaseinstance
Available
Instance

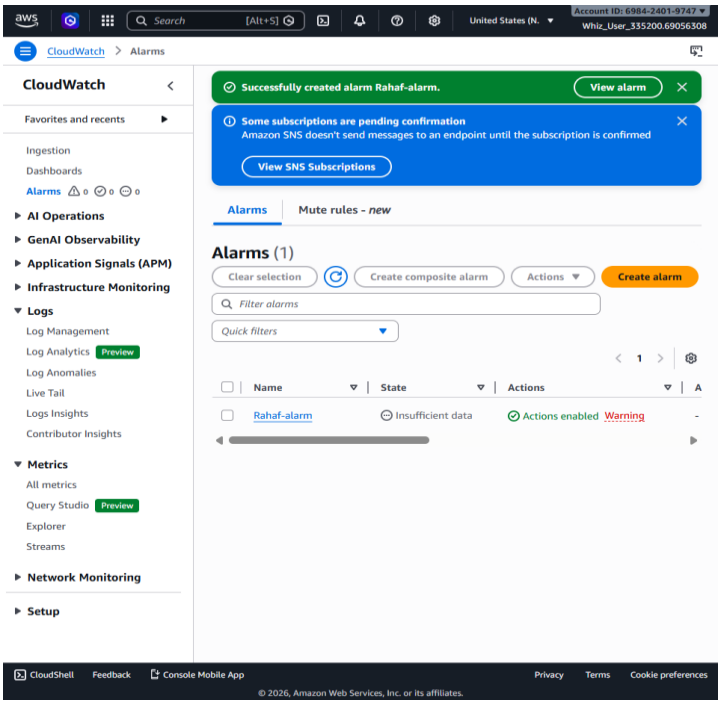
6. Security

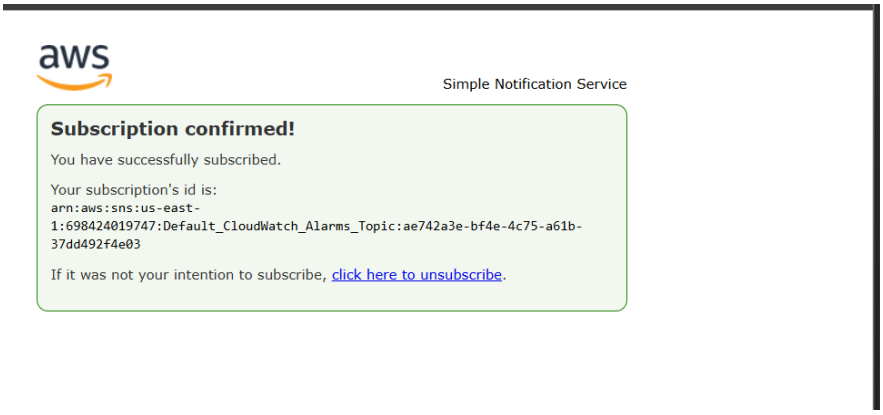
Here I create a IAM Roles

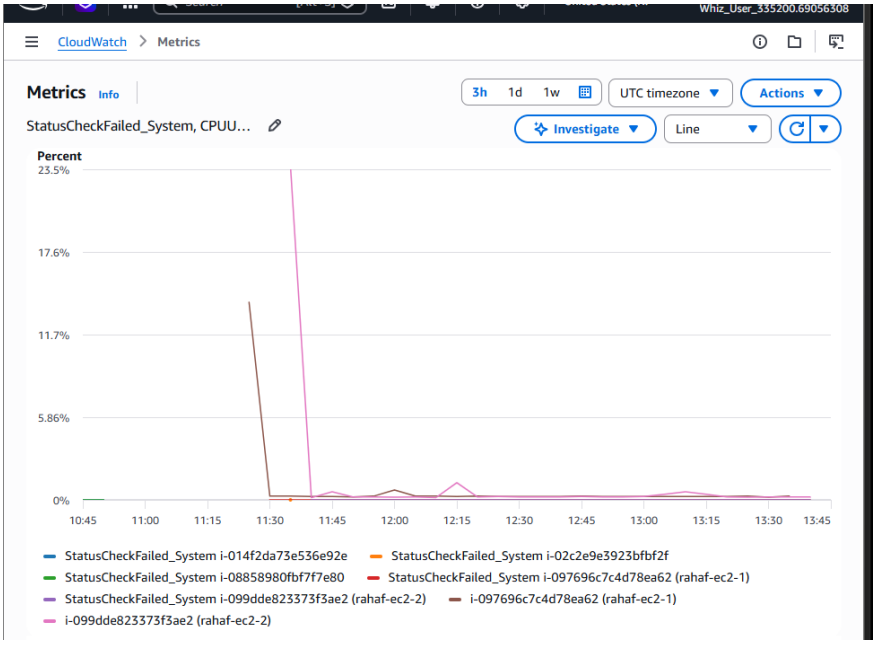


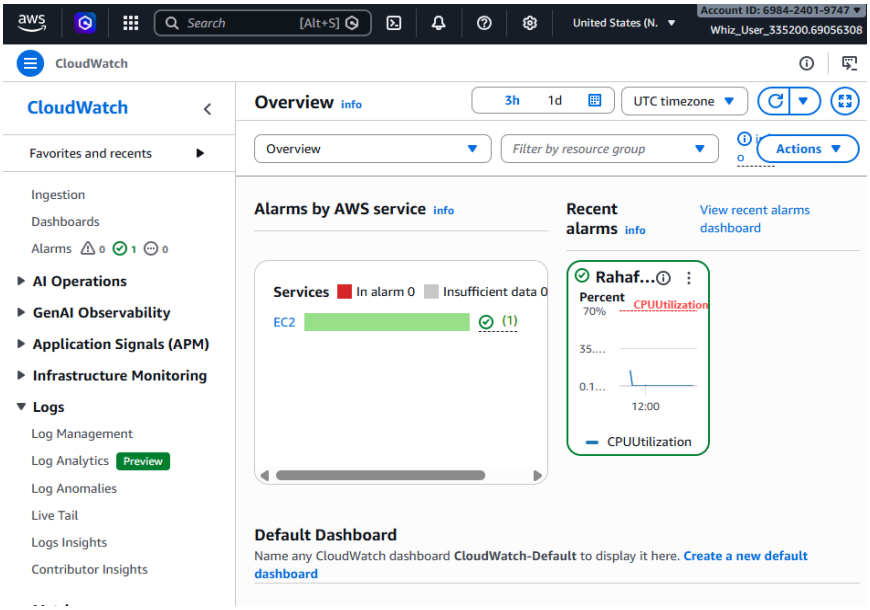


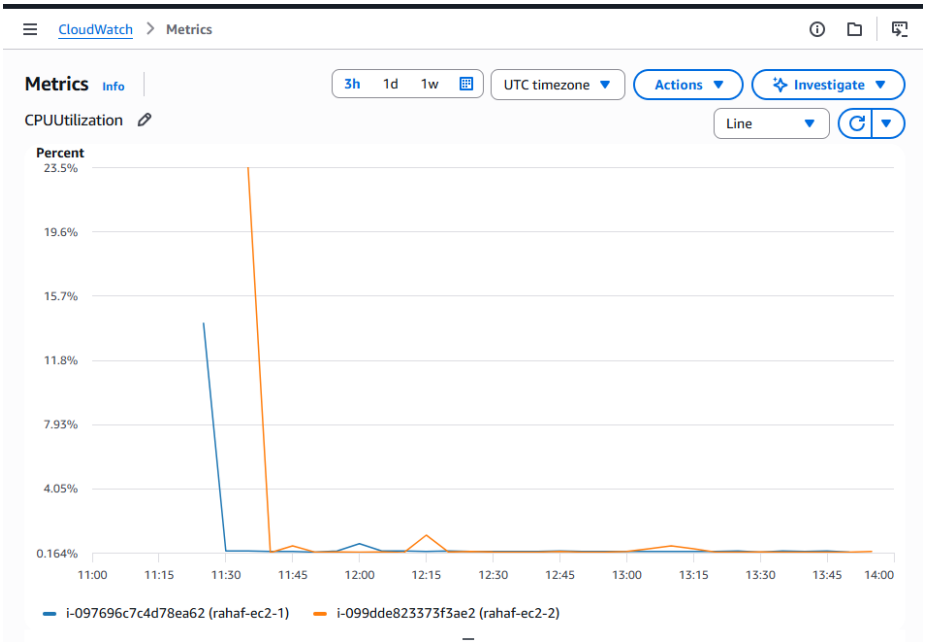
Here I create an Alarms






Whiz_User_335200.69056308
CloudWatch > Metrics
Metrics Info
3h 1d 1w
UTC timezone
Actions
StatusCheckFailed_System, CPUU...
Investigate
Line
Percent
23.5%
17.6%
11.7%
5.86%
0%
10:45 11:00 11:15 11:30 11:45 12:00 12:15 12:30 12:45 13:00 13:15 13:30 13:45
StatusCheckFailed_System i-014f2da73e536e92e
StatusCheckFailed_System i-02c2e9e3923bfbf2f
StatusCheckFailed_System i-08858980fbf7f7e80
StatusCheckFailed_System i-097696c7c4d78ea62 (rahaf-ec2-1)
StatusCheckFailed_System i-099dde823373f3ae2 (rahaf-ec2-2)
i-097696c7c4d78ea62 (rahaf-ec2-1)
i-099dde823373f3ae2 (rahaf-ec2-2)


aws
Search [Alt+S]
United States (N.
Account ID: 6984-2401-9747
Whiz_User_335200.69056308
CloudWatch
CloudWatch
Favorites and recents
Ingestion
Dashboards
Alarms 0 1 0
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management
Log Analytics Preview
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Overview info
3h 1d
UTC timezone
Overview
Filter by resource group
Actions
Alarms by AWS service info
Services In alarm 0 Insufficient data 0
EC2 (1)
Recent alarms info
View recent alarms dashboard
Rahaf...
Percent
70%
CPUUtilization
35....
0.1...
12:00
CPUUtilization
Default Dashboard
Name any CloudWatch dashboard CloudWatch-Default to display it here. Create a new default dashboard


CloudWatch
Metrics
Metrics
Info
3h
1d
1w
UTC timezone
Actions
Investigate
CPUUtilization
Line
Percent
23.5%
19.6%
15.7%
11.8%
7.93%
4.05%
0.164%
11:00
11:15
11:30
11:45
12:00
12:15
12:30
12:45
13:00
13:15
13:30
13:45
14:00
i-097696c7c4d78ea62 (rahaf-ec2-1)
i-099dde823373f3ae2 (rahaf-ec2-2)